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Designer Peptide Dendrons and Dendrimers Based Soft Materials Through Self-Assembly

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A series of new aspartic and glutamic acid based peptide dendrons and dendrimers were designed and synthesized. The supramolecular assembly of these novel molecules were studied by scanning electron microscopy (SEM), atomic force microscopy (AFM), and attenuated total reflectance Fourier transform infrared (ATR-FTIR) spectroscopy. The dendrons and dendrimers formed gel in aqueous or in organic phase with excellent storage modulus as measured by rheometry. Glutamic acid-based higher generation dendrons and dendrimers showed a minimum gelation concentration as low as 1 wt/v (%) in $CHCl_3$. Interestingly, the higher generation dendrimers **5d**, **6d**, and **7d** gelated even polar solvent DMSO. The lipidated peptide dendrons showed thermotropic liquid crystalline behavior with the formation of columnar and smectic phase as confirmed by differential scanning calorimetry (DSC) and polarization optical microscopy (POM). These lipidated peptides exhibited well-ordered specific modes of self-assembly and hence could be exploited for several applications such as drug delivery vehicles and in tissue engineering.

Introduction

Molecular self-assembly is a powerful strategy for the construction of functional materials.[1] Self-assembly of small molecules to organogels is a vigorously pursued area, since organogels are useful as drug delivery systems, biomaterials, and templates for fabricating nanostructures.[2] Polypeptide-based materials have several applications due to their well-defined nano-and microstructures.[3] Peptides are the simplest class of self-assembling molecules that can be employed as biomaterials.[4] Peptides are readily accessible by well-established peptide synthesis protocols. Peptide amphiphiles (PAs) have recently received significant interest as gel forming biomaterials owing to their strong self-assembling nature.[5] PAs have an N-terminal alkyl tail, which drives self-assembly. Peptides containing di-phenylalanine (FF)[6] or fluorenylmethoxycarbonyl (Fmoc) units have also showed strong self-assembling behavior due to the π-π stacking of the aromatic groups.[7] Short Fmoc-FF peptides spontaneously assemble to produce gels due to π-π stacking by the Fmoc groups.[8]

Amphiphilic dendrons and dendrimers with several peptide bonds have the potential to organize into the hydrogen-bonded supramolecular assembly.[9] Due to their enormous surface area and the branched architecture, they can immobilize the solvent and hence can act as potential gelators. These dendritic gels are of great interest because of their applications in material and biological sciences.[10–11] However, despite their applications, the self-assembly of peptide-based dendrimers are less explored.

In this paper, we intend to combine the peptides and the lipid units for the synthesis of a novel class of dendrimers. These lipidated peptide dendrimers are expected to show interesting self-assembling properties, owing to their branched architecture, peptide bonds and lipid chains.

Results and Discussion

Design, Synthesis and characterization

Synthetic lipidated amphiphiles have high tendency to self-assemble into various nano structures, hence can be used for several biological applications.[12–13] We have attached hydrophobic lipid chain to amino acid by the carboxylic acid functionalization. These lipidated molecules were further connected to another dendritic unit; bearing large number of surface groups to form Janus dendrimers.[14] The hydrophobic lipid chains and higher surface area of these designer dendritic molecules can self-assemble owing to the presence of a large number of hydrogen bond donors and acceptors. Although Lys-based dendrons are very well-known to gelate different solvent systems but incorporation of lipidated units into it will improve not only their gelation but also facilitate self-assembly.[15]

All the dendrimers were synthesized in a convergent synthetic strategy. The alkyne truncated Lys dendrons were reacted with an azide truncated Asp/Glu dendrons to obtain unsymmetrical dendritic architectures.[16] We chose Lys and Asp/Glu amino acid residues for the synthesis of dendrimers, since they possess easily functionalizable side chains. The Lys-based dendrons were equipped with an alkyne at the C-terminus and Asp/Glu dendrons were functionalized with an azide moiety at

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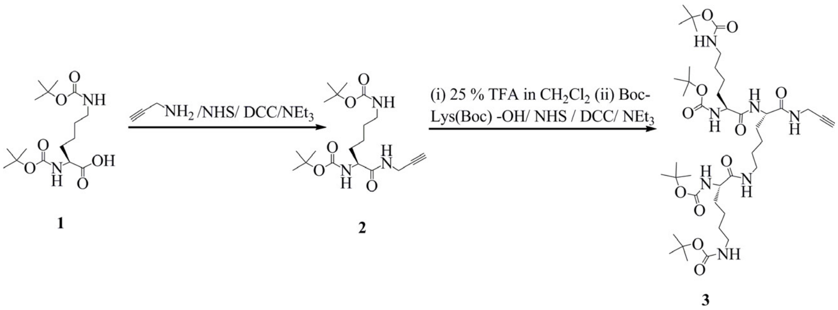
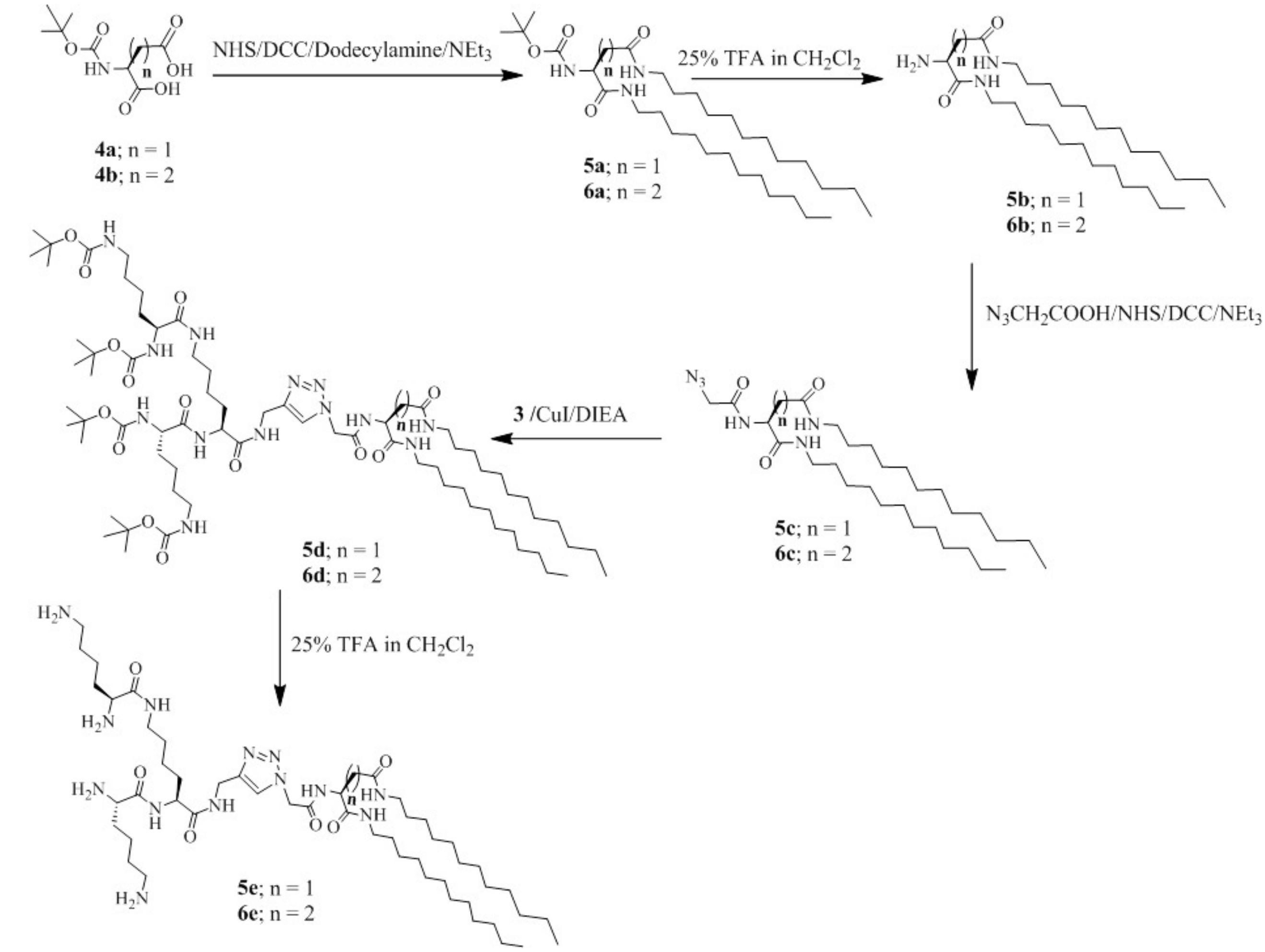


Scheme 1. Preparation of alkyne truncated Lys dendron.

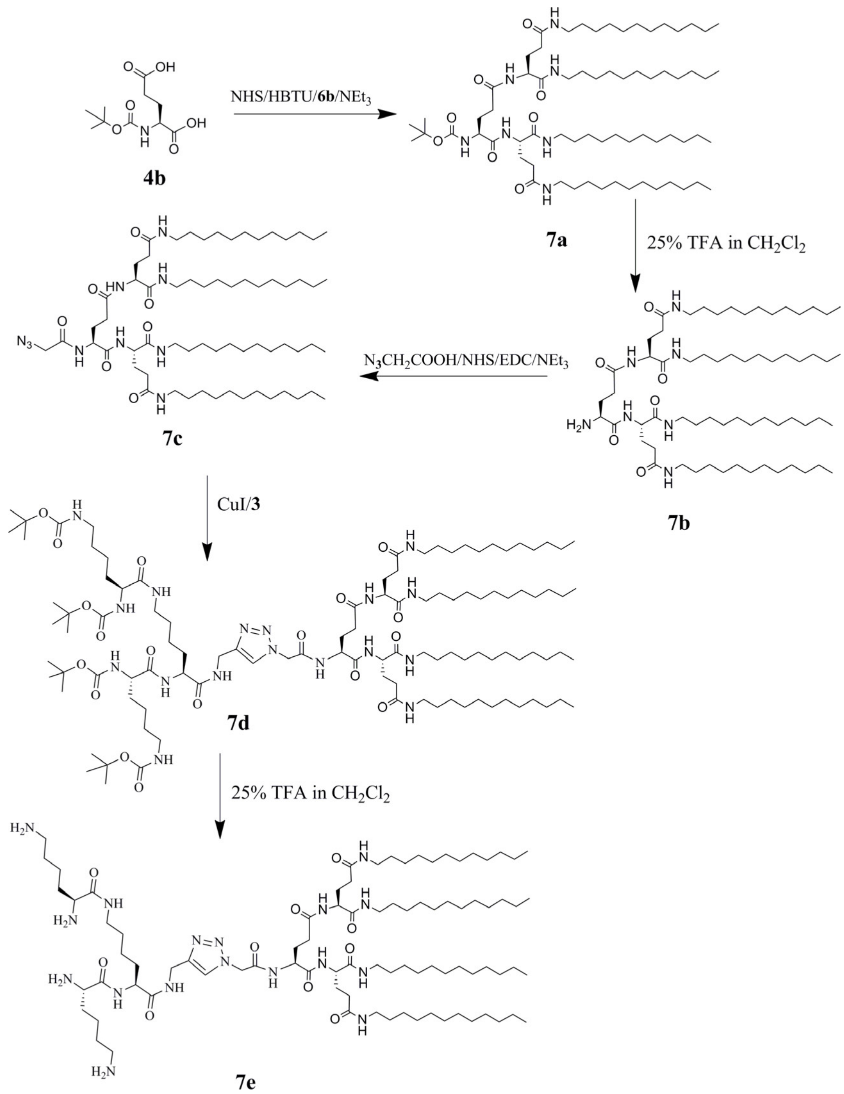


Scheme 2. Synthesis of Asp/Glu based lipidated dendrons and dendrimers.

the N-terminus. Lysine was chosen as the building block for the synthesis of dendrimers, since it has α and ε amino groups and hence serves as branching units. Similarly, Asp/Glu provide additional carboxylic acid groups.

Bis-Boc protected lysine **1**, was coupled with propargylamine by using dicyclohexylcarbodiimide (DCC) as a coupling agent and N-hydroxysuccinimide (NHS) as an activator to afford alkyne derivatized lysine **2.** Compound **2** after deprotection and further coupling with bis-Boc protected Lys gave second generation dendron **3** (Scheme 1).[17]

In order to synthesize lipidated dendrons, N-protected Asp/Glu (**4a**/**b**) was reacted with dodecylamine to get lipidated compounds **5a** and **6a.** These dendrons were converted to the azide truncated dendrons **5c** and **6c**, by reacting azidoacetic acid with N-deprotected lipidated **5b** and **6b** dendrons (Scheme 2). The lipidated intermediate **6a** was treated with trifluoroacetic acid to produce dendron **6b**, which upon reaction with Boc-Glu-OH (**4b**) generated lipidated dendron **7a.** Further deprotection afforded dendron **7b**, which upon reaction with azidoacetic acid produced the azide truncated dendron **7c**



Scheme 3. Synthesis of higher generation dendrons (**7 a**-**7 c**) and dendrimers (**7 d**-**e**) from Boc-protected glutamic acid.

(Scheme 3). The two sets of dendrons, one equipped with alkyne **3**, and other truncated with azide **5 c**, **6 c** and **7 c** were coupled by a chemoselective Cu(I) catalyzed 1,3-dipolar cycloaddition reaction to give unsymmetrical dendrimers **5 d**-**e**, **6 d**-**e** and **7 d**-**e** (Schemes 2 and 3). The Cu(I) catalyzed azide-alkyne cycloaddition reaction provided full-size dendrimers in good yields.[18]

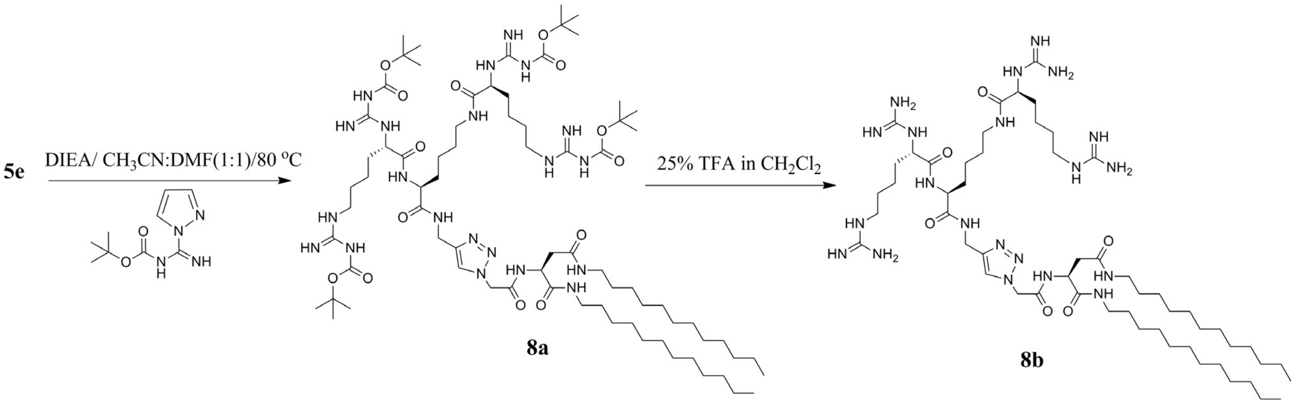
Apart from the dendrimers having amino groups on the surface, we also synthesized guanidine dendrimers **8 a**-**b.** The amine containing dendrimer **5 e** can easily be converted to guanidine by reacting it with 1-H-Pyrazole-(N-tert-butoxy carbonyl) carboxamidine (Scheme 4). All of the dendrons and dendrimers were characterized by ^{1}H NMR and ESI mass spectrometry.

Investigation of self–assembly

Most of the protected dendrons (**5 a**, **5 c**, **6 c**, and **7 a**-**7 c**) and dendrimers (**5 d**, **6 d**, and **7 d**) showed ready gelation in organic solvents (Figure 1 A). Peptide dendrimer gel is formed, when non-polar solvent (n-hexane) is added to a homogeneous solution of dendrimer in a polar solvent (chloroform). The morphology of the gels was examined by scanning electron microscopy (SEM) and by atomic force microscopy (AFM). All the gel samples were prepared from appropriate solvent systems (Table S1), and the xerogels were used for the microscopic analysis.

Most of the dendrons (**5 a**, **5 c**, and **6 c**) formed ready gel formation in 1:1 $CHCl_3$/Hexane mixture. The lipidated dendrons **7 a**-**7 c** showed gelation in chloroform (Figure 1 A).

The dendrimers **5 d**, **6 d**, and **7 d** formed gels even in a highly polar solvent like dimethyl sulfoxide (DMSO, Figure S1). Solvent-like DMSO is known to break intermolecular interactions between the molecules due to the strong tendency of DMSO to form intermolecular hydrogen bonds with the compound. The formation of gels in DMSO indicates a high self-assembling property of the lipidated dendrimers **5 d**, **6 d**, and **7 d.** SEM images of xerogels from **5 d**, **6 d** and **7 d** showed thick sausage like morphology (Figure S2). The deprotected dendrimers **5 e**, **6 e**, and **7 e** were found to be non-gelating in nature. The gua-



Scheme 4. Synthesis of guanidine-based dendrimers **8a**-**8b**.

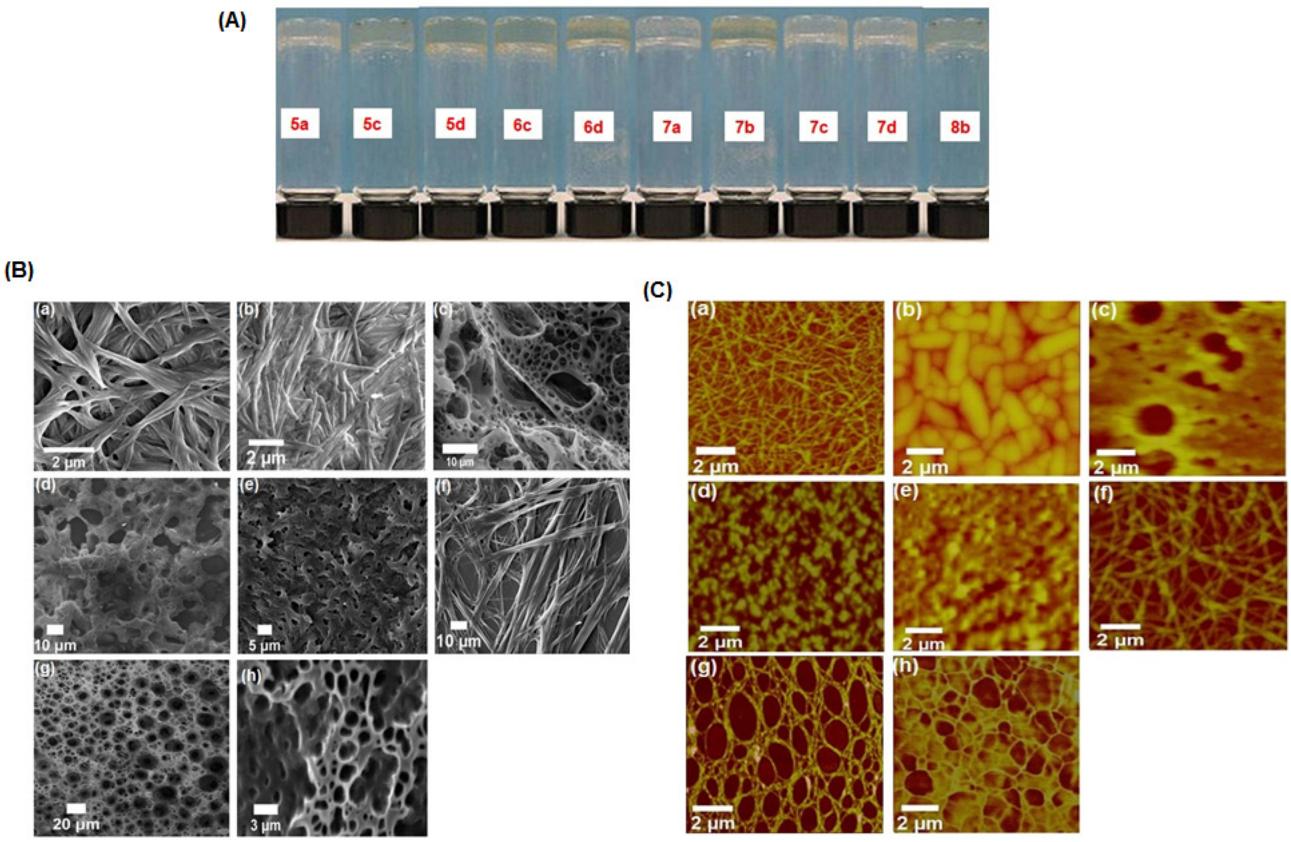


Figure 1. Panel (A) Photographs of inverted vials Containing gels. Panel (B) SEM images of the xerogels (a) **5a** (b) **5c** (c) **5d** (d) **6c** (e) **6d** (f) **7a** (g) **7b** (h) **7c.** Panel (C) AFM images of the xerogels (a) **5a** (b) **5c** (c) **5d** (d) **6c** (e) **6d** (f) **7a** (g) **7b** (h) **7c.** Gel formation was observed in 1:1 $CHCl_3$/Hexane for **5a**, **5c**, **6c**, **5d**, **6d** and **7d**, whereas gel formation was observed in $CHCl_3$ for **7a**-**7c**.

nidinylated dendrimer **8b** could gelate water to form a thick hydrogel at 11 mM concentration (Figure 1A).

SEM images of xerogels of **5a** and **7a** showed fibers. The average diameter of fibers of **5a** was 0.26 µm and 4.3 µm for **7a** with lengths of several micrometers (Figure 1B). These gels showed entangled and extensive networks of fibers. AFM imaging also supported the above morphology and the dimensions of fibers were comparable in both SEM and AFM (Figure 1C). The SEM imaging revealed that dendron **5c** assembled to form a stick-like morphology. They have an average thickness of 0.65 µm and length of 8.1 µm. AFM imaging also supported this observation (Figure 1C). The dendrons **6c**, **7b** and **7c** exhibited porous surface morphology. In the case of SEM images of **7b**, the pores are of ~3.9 µm diameter, whereas for **6c** and **7c** the pores have diameter of ~1.7 µm (Figure 1B). AFM imaging also supported the microporous nature of these dendrons (**6c**, **7b**, and **7c**) (Figure 1C). The SEM and AFM images of xerogels from dendrimers **5d** and **6d** revealed microporous surface morphology (Figures 1B–C). SEM images of **7d** and **8b** showed thick sausage-like morphology (Figure S2).

Attenuated total reflectance (ATR-FTIR) spectroscopic studies

The strong self-assembly of lipidated dendrons and dendrimers results in the gelation. The mode of self-assembly could be studied by ATR-FTIR spectroscopy.[19] The peptide dendrimer with multiple amide bonds could involve in extensive intermolecular hydrogen bonding interactions. Therefore, the ATR-FTIR of gels is expected to be different from solid samples. The amide NH-stretching region around 3500–3200 and the carbonyl stretching region1750-1600 cm^{-1} will provide information about the nature of hydrogen bonding. The strong intermolecular hydrogen bonding interactions will decrease the stretching vibrational frequencies of the amide NH and the carbonyl stretching since these are primarily involved in the hydrogen-bonded assembly in the gel state. [20] The ATR-FTIR of **5a** showed bands at 3345 and 3320 cm^{-1} corresponding to the amide -NH- stretching and amide I and II vibrations at 1687, 1549 cm^{-1} (Figure S3, Table 1). The xerogel of **5a** showed bands at 3308 cm^{-1} (NH) and at 1683, 1520 cm^{-1} (amide I and II). The decreased stretching vibrational frequencies in the gel indicate hydrogen bond networks in the gel state (Figure S3). The ATR-FTIR of azide truncated dendron **5c** in the gel state showed a considerable decrease in the stretching frequencies of amide NH and the amide I and II bands (Table 1, Figure S3) compared to the solid samples. The higher generation azide truncated dendrons **6c** and **7c** showed a similar trend.

Table 1. ATR-FTIR spectroscopic data for the dendrons and dendrimers

Dendrons and Dendrimers	Stretching frequency for amide -NH cm^{-1}		Stretching frequency for amide I & II (carbonyl) cm^{-1}	
	Solid	Gel	Solid	Gel
5a	3345	3308	1687, 1549	1683, 1520
5c	3283	3278	1641, 1554	1637, 1549
5d	3287	3278	1683, 1641	1678, 1633
6c	3288	3274	1645, 1562	1641, 1549
6d	3283	3278	1687, 1549	1678, 1520
7a	3345	3287	1687, 1522	1683, 1516
7b	3291	3283	1691, 1520	1683, 1518
7c	3287	3278	1641, 1558	1633, 1541
7d	3287	3274	1645, 1521	1638,1516

The dendrimer **5d** in the gel state showed NH and amide I and II bands at a lower frequency compared to the solid samples. Similarly, the ATR-FTIR of gel sample from dendrimer **6d** showed decreased amide NH and carbonyl vibrational frequencies supporting the hydrogen bonding responsible for gelation (Table 1, Figure S3).

The higher generation lipidated dendrons **7a**-**7c** and dendrimer **7d** could form very strong and thick gel in chloroform. The dendrons **7a** and **7b** showed differences in the ATR-FTIR of gel and the solid sample. ATR-IR of hydrogel from guanidinylated dendrimer **8b** showed considerable differences in the amide NH and CO vibrations (Table 1, Figure S3).

Rheological studies

Although, gels have many potential applications in food industry, cosmetics and in pharmaceuticals, but their poor mechanical strength, severely restrict their applications. Gelation of organic solvents by dendrons and dendrimers is caused by the self-assembling nature of molecules to form various network structures, which in turn depends on their chemical structures.[21]

In order to find out the viscoelastic properties of these organogels, we have performed rheological measurements. Viscoelastic moduli of the gel can predict the nature and strength of the gel. Storage modulus (G′) gives the magnitude of the solid-like behavior of the gel (elastic nature) and loss modulus (G“) indicates the viscous (liquid like) property of the gel. For a typical gel, G′ should be greater than G″ and both should be independent of the applied frequency.[22] The rheological measurements were made to quantify the tolerance performance of the gel to external forces. Initially, amplitude strain experiments were performed at a constant frequency (1 Hz) for evaluating the linear viscoelastic behavior. The variation of G′ and G″ were monitored as a function of strain sweep and it was found that they were independent of strain (1 to 5% strain (Figure S4). The frequency sweep experiments were performed under a constant strain of 1%. The G′ and G″ values of **5a** and **5c** gels were independent of applied frequency (Figure 2). The dominant elastic behavior of **5a** and **5c** gels is evident from the higher value of G′ compared to G” (Table 2). The ratio of G′/ G“ for **5a** (2.01) and **5c** (1.71) indicated that **5a** forms a rigid gel compared to **5c.** The reason could be the N-terminal groups on these dendrons, which establish extensive noncovalent interaction and self-assembly in these organogels. The gel of **5c** re-

Table 2. Gelation ability and mechanical strength of dendrons and dendrimers.

Dendrons and Dendrimers	Wt/v (%)	Solvent system	G′/G″
5a	12.19	$CHCl_3$/Hexane (1:1)	2.01
5c	12.49	$CHCl_3$/Hexane (1:1)	1.71
6c	12.49	$CHCl_3$/Hexane (1:1)	2.47
7a	0.35	$CHCl_3$	4.69
7b	0.79	$CHCl_3$	1.67
7c	0.54	$CHCl_3$	3.01
5d	1.04	$CHCl_3$/Hexane (1.1)	1.95
6d	1.72	$CHCl_3$/Hexane (1:1)	2.61
7d	0.61	$CHCl_3$/Hexane (1:1)	6.53
8b	12.73	H_2O	2.57
5d	1.01	DMSO	1.22
6d	2.20	DMSO	2.71
7d	2.19	DMSO	6.87

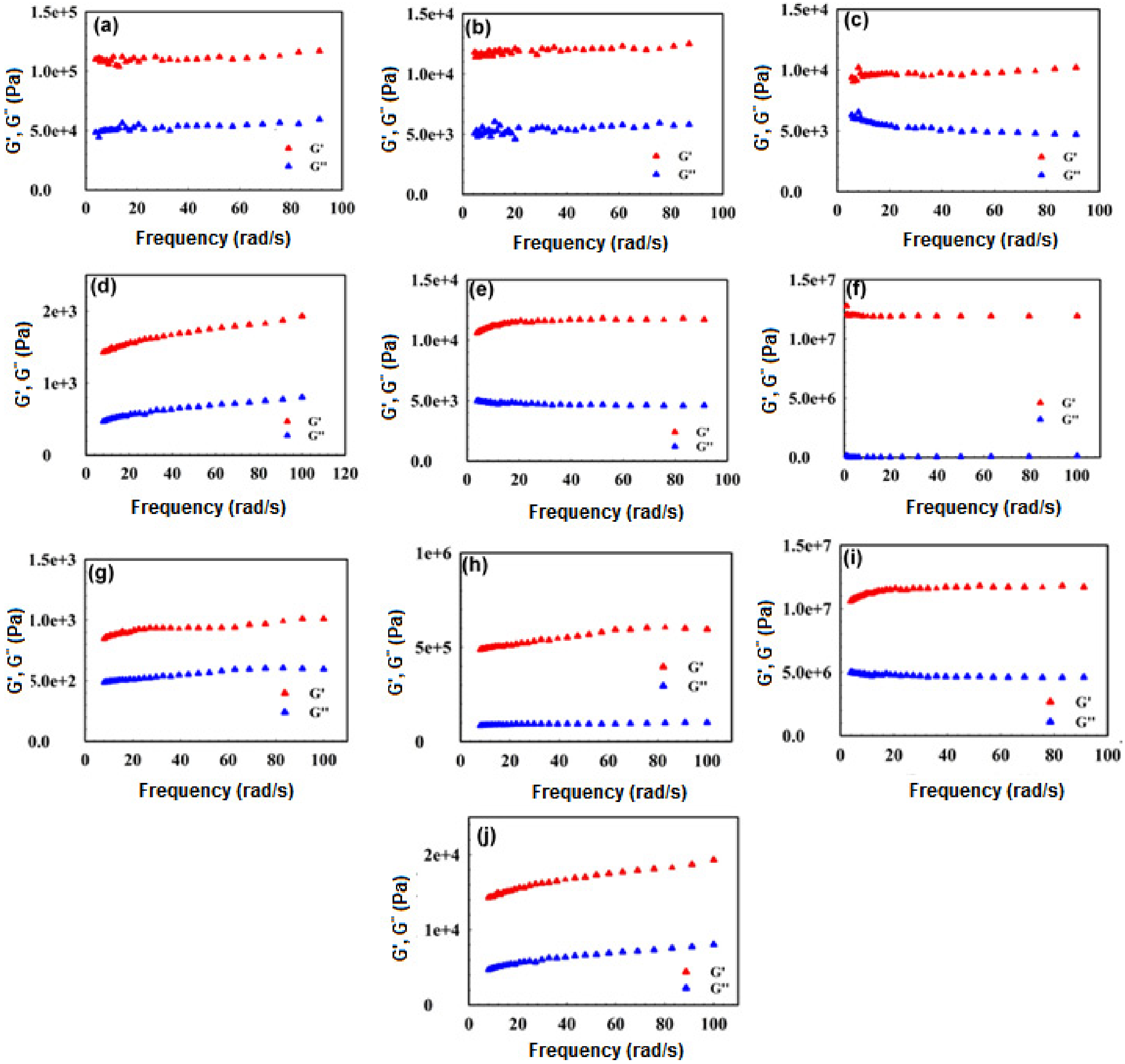


Figure 2. Frequency sweep measurements of the organogels from (a) **5a** (b) **5c** (c) **5d** (d) **6c** (e) **6d** (f) **7a** (g) **7b** (h) **7c** (i) **7d** (j) **8b.** Gels were studied at 25 °C. Gel formation was observed in 1:1 $CHCl_3$/Hexane for **5a**, **5c**, **6c**, **5d**, **6d** and **7d**, whereas gel formation was observed in $CHCl_3$ for **7a**-**7c** and **8b** formed gel in water. G′ and G″ stands for storage and loss modulus.

tained its elastic nature up to a shear stress of 15 Pa, while above the critical stress value of 20 Pa, both G′ and G″ abruptly fell to a very low value (Figure S4).

For the organogel from the dendrimer **5d**, G′ is greater than G″ indicating a high elastic modulus (solid-like behavior) of gel. The dynamic moduli of the **5d** gel showed that over a range of applied stress, both G′ and G″ were invariant indicating the rigidity (Figure S4). The ratio of G′/G″ (1.95) indicates a high elastic nature for this dendrimer gel (Figure 2).

The rheomteric studies on the gels from Glu-based dendron **6c** and dendrimer **6d** revealed that G′ and G″ are virtually independent of applied frequency (Figure 2). The ratio of G′/G″ for **6c** (2.47) and **6d** (2.61) indicated a solid-like nature for the gels (Table 2).

In the case of **7a**-**7c,** G′ > G″ in the entire frequency range (Table 2) revealing high elastic modulus for these gels indicative of strong gel (Figure 2, Table 2). The high rigidity of the **7d** gel is evident from the higher G′/G″ (6.87) value (Table 2). The rheometric analysis also revealed that the guanidinylated dendrimer **8b** formed (Figure 2) a rigid hydrogel (G′/G″ = 2.57).

Rheological measurements for the dendrimer gels from **5d**, **6d** and **7d** in DMSO also showed a high elastic nature as the ratio of G′/G″ was found to be 1.22, 2.71 and 6.87 respectively (Table 2. Figure S4).

Investigation of the liquid crystalline behavior of dendrons

Peptides containing many amino acids can establish extensive inter and intramolecular non-covalent interaction to from excellent self-assembled supramolecular structures. Studies showed that these self-assembled structures exhibited both thermotropic liquid crystalline behavior (temperature induced) and gelation behavior (solvent induced). Percec and co-workers designed and synthesized a series of dendrons carrying alkyl chains with thermotropic liquid-crystalline properties.[23] Stupp *et al.* reported liquid crystalline dendritic molecules.[24] However, liquid crystalline peptide dendrimers are scarce. In this work, we have studied the liquid crystalline properties of the designer lipidated peptide dendrons. Aspartic and glutamic acids could be easily transformed to dendritic molecules. Such dendritic molecules with long chains are expected to exhibit liquid crystalline phase behavior.

Thermotropic liquid crystal properties of these lipidated dendrons were studied by differential scanning calorimetry (DSC) in combination with polarization optical microscopy (POM) and the details are presented in Table 3. Interestingly, all of the lipidated Asp and Glu-based lipidated dendrons showed smectic phase. The DSC thermogram of N-Boc protected bis-lipidated Aspartic acid **5 a** showed two phase transitions at 90 °C (crystalline to smectic phase) and an isotropic phase at 135 °C (Figure 3 a). Upon cooling from 135 °C to 90 °C, a clear smectic phase with a columnar texture was observed in the POM. The **5 b** (N-deprotected derivative of **5 a**) showed two phase transitions at 90 °C and at 70 °C (Figure 3 b). A focal conic fan-shaped texture (Figure 3 d) upon cooling from the isotropic phase (70 °C) is observed. The DSC scan of azide truncated dendron **5 c** showed two clear phase transitions; one at 104 °C and another at 180 °C (Figure 3 c). The POM images showed a tetragonal columnar phase with extended toothed fan type smectic texture (Figure 3 e) with fibres growing in one direction.

Table 3. Transition temperatures determined by DSC and POM

Compounds	Phase (DSC)	Phase (POM)	Phase
5 a	Cr (90.3 °C)	Sm (90 °C)	I (135 °C)
5 b	Cr (70.8 °C)	Sm (70 °C)	I (90 °C)
5 c	Cr (90.2 °C)	Sm (100 °C)	I (180 °C)
6 a	Cr (70.5 °C)	Sm (70 °C)	I (110 °C)
6 b	Cr (115.9 °C)	Sm (115 °C)	I (140 °C)
6 c	Cr (120.4 °C)	Sm (120 °C)	I (180 °C)
7 a	Cr (130.9 °C)	Sm (130 °C)	I (160 °C)
7 b	Cr (130.4 °C)	Sm (130 °C)	I (160 °C)

DSC scanning rate 10 °C/min; Cr: crystalline phase; I: Isotropic melt, sm: Smectic phase

Like the Asp derivatives **5 a-c**, Glu-derivatives **6 a-c** also showed LC phases. The Glu derivative **6 a** showed two distinct phase transitions in DSC at 70 °C and 110 °C (Figure 4 a) with a smectic columnar phase (Figure 4 f) upon cooling from 110 °C to 30 °C. The N-deprotected derivative **6 b** showed a battonet smectic phase (Figure 4 g) upon cooling from isotropic point to 40 °C. DSC thermogram supported the observed transition temperatures (Figure 4 b). The azide truncated dendron **6 c**, showed a focal conic flower-like texture upon cooling from 180 °C to 120 °C (Figure 4 c, Figure 4 h).

Higher lipidated dendron **7 a** showed broken fan-like texture in POM upon cooling from 164 °C to 132 °C (Figures 4d and 4i). The dendron **7 b** showed smectic focal fan-like texture with an isotropic melt at 164 °C with a crystallization temperature at 132 °C (Figures 4e and 4j). The DSC thermogram clearly depicts this fact.

Conclusions

In conclusion, we have designed and synthesized a series of Asp/Glu based lipidated peptide dendrons and dendrimers. The Cu-catalysed click chemistry allowed us to link different dendrons to obtain Janus dendrimers in good yields. These lipidated peptide dendrimers showed excellent gelation either in organic solvents or in water. A detailed ultramicroscopic analysis showed a diverse range of morphologies. Interestingly, the lipidated dendrons also exhibited thermotropic liquid crystalline smectic phase.

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Keywords: Dendrimers · liquid crystals · peptides · self-assembly

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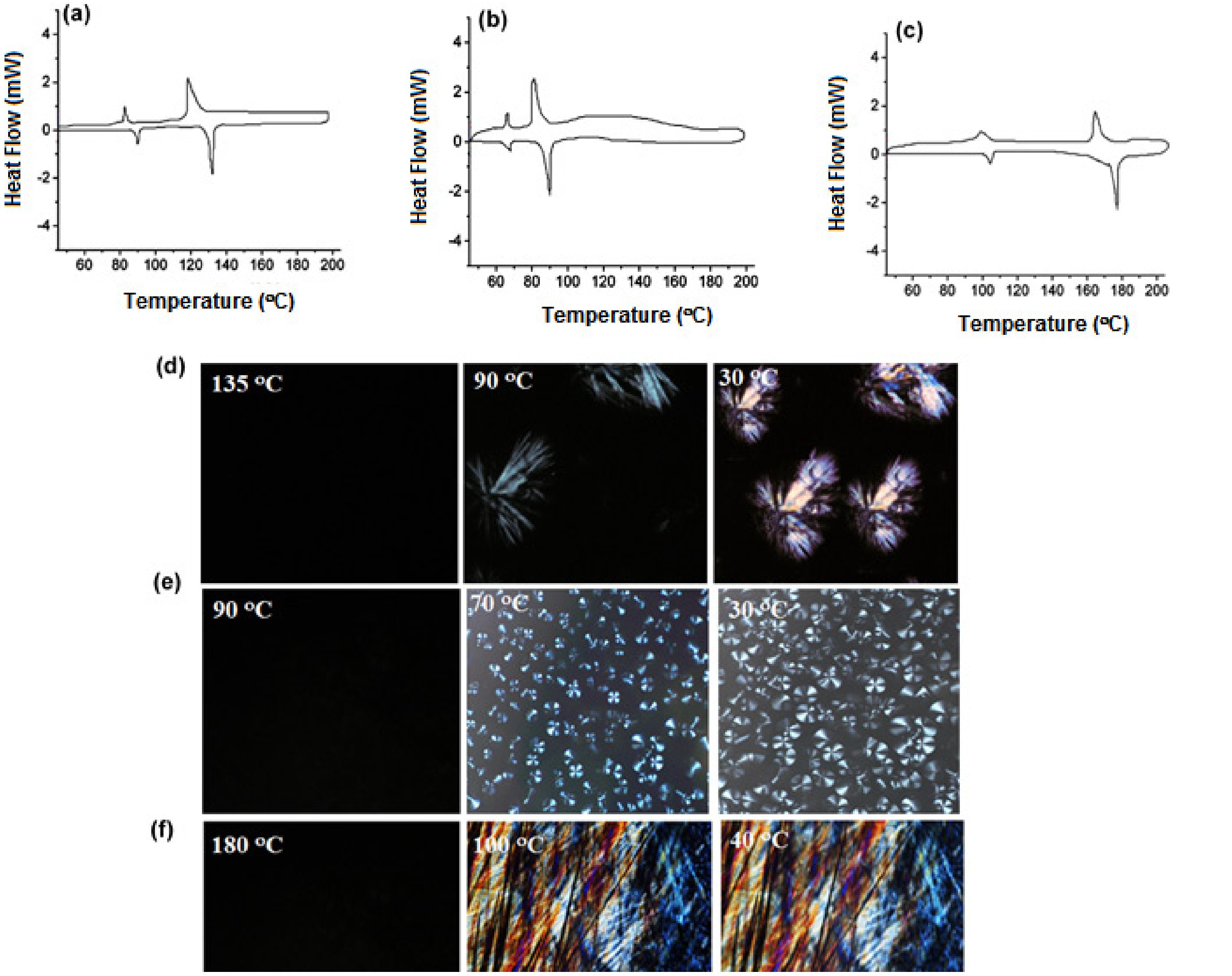

Figure 3. (a-c) represents the DSC thermograms for **5 a**-**5 c** respectively and (d-f) represents POM images at different temperatures for **5 a**-**5 c.**

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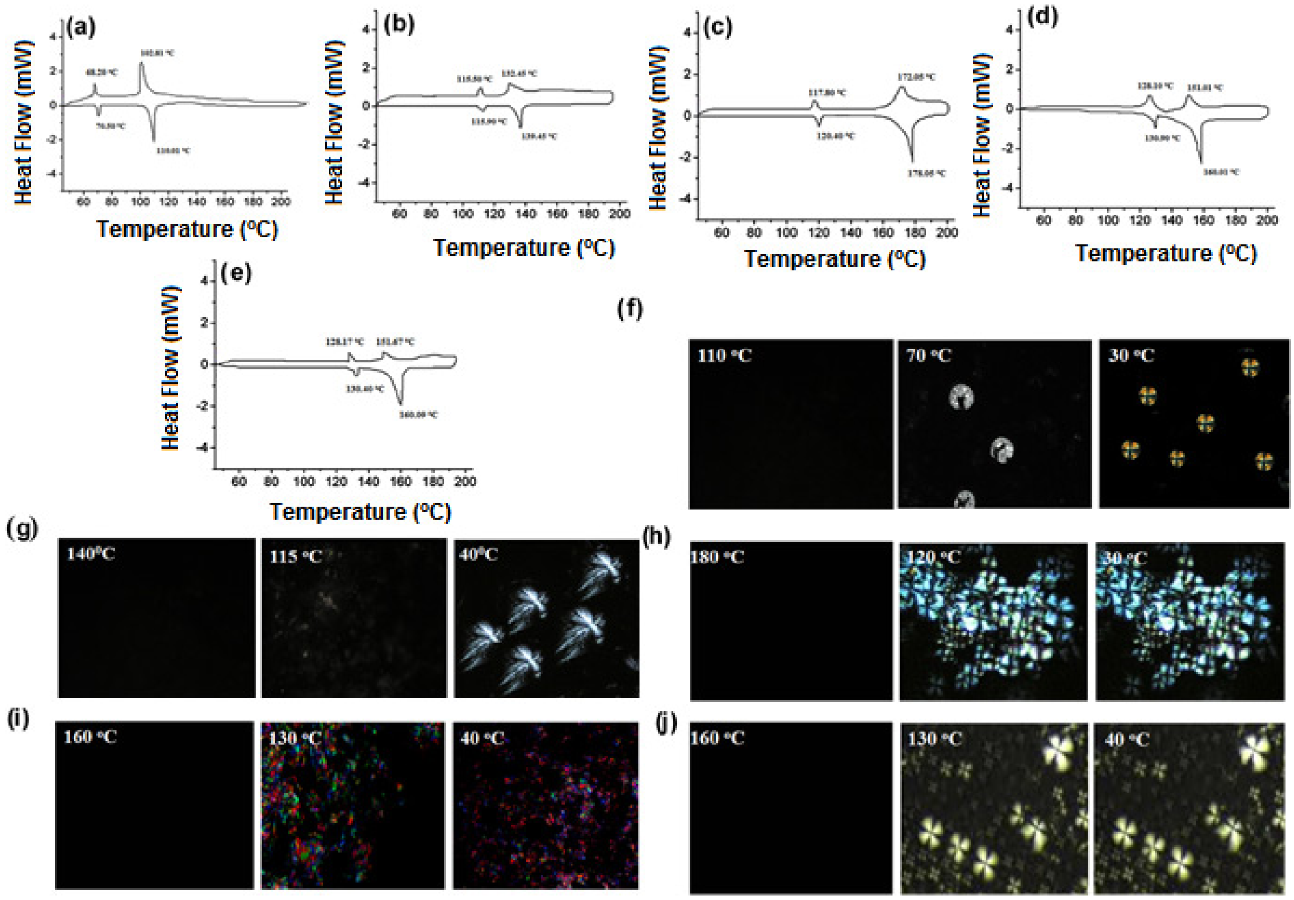

Figure 4. (a-e) represents the DSC thermograms for **6 a-6 c** and **7 a-7 b** respectively and (f-j) represents POM images at different temperatures for **6 a-6 c** and **7 a-7 b.**

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